Notes on Gauge Theory

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4 Characteristic Classes

Let $\pi : E \to M$ be a rank r vector bundle, ∇ be a connection on E. Its curvature $R^\nabla := \nabla^2$ is an $\mathrm{End}(E)$-valued 2-form on M. Under a local trivialization $\{(U_\alpha, \psi_\alpha)\}$ of E, ∇ can be represented by an $r \times r$ matrix of 1-forms ω_α, and R^∇ can be represented by an $r \times r$ matrix of 2-forms Ω_α.

The main goal of this section is the following:

1. An invariant polynomial Φ on $\mathfrak{gl}(r, \mathbb{R})$ determines a global form $\Phi(R^\nabla)$ on M.
2. The global form $\Phi(R^\nabla)$ is closed.
3. The class $[\Phi(R^\nabla)]$ is independent of the choice of ∇, thus defines an invariant of E.
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4.1 Chern-Weil Theory

Repeat the settings above: Let $\pi : E \to M$ be a rank r vector bundle, ∇ a connection on E. The curvature R^∇ is an $\mathrm{End}(E)$-valued 2-form on M. Under a local trivialization $\{(U_\alpha, \psi_\alpha)\}$ of E, ∇ can be represented by an $r \times r$ matrix of 1-forms ω_α, and R^∇ can be represented by an $r \times r$ matrix of 2-forms Ω_α.

Recall that the curvature form Ω_α is defined by

$$\Omega_\alpha = d\omega_\alpha + \omega_\alpha \wedge \omega_\alpha.$$

It satisfies the second Bianchi identity

$$d\Omega_\alpha = \Omega_\alpha \wedge \omega_\alpha - \omega_\alpha \wedge \Omega_\alpha,$$

and the general Bianchi identity

$$d\Omega_\alpha^k = \Omega_\alpha^k \wedge \omega_\alpha - \omega_\alpha \wedge \Omega_\alpha^k.$$

The transformation formula for ω_α and Ω_α is given by

$$\omega_\beta = g_{\alpha\beta}^{-1} \omega_\alpha g_{\alpha\beta} + g_{\alpha\beta}^{-1} dg_{\alpha\beta}, \qquad \Omega_\beta = g_{\alpha\beta}^{-1} \Omega_\alpha g_{\alpha\beta}.$$

where $g_{\alpha\beta}$ is the transition function.

Definition 4.1. Let $X = (x^i_j)$ be an $r \times r$ matrix with indeterminate entries. A polynomial $\Phi(X)$ on $\mathfrak{gl}(r, \mathbb{R}) = \mathbb{R}^{r\times r}$ is a polynomial in the variables x^i_j. A polynomial $\Phi(X)$ on $\mathfrak{gl}(r, \mathbb{R})$ is Ad**-invariant (or simply invariant)** if

$$\Phi(A^{-1} X A) = \Phi(X), \qquad \forall\, A \in \mathrm{GL}(r, \mathbb{R}). \tag{4.1}$$

Remark. If X is a matrix of real numbers, then $\Phi(X)$ is a real number. If X is a matrix of 2-forms, then $\Phi(X)$ is also a differential form.

Fact 1. If (4.1) holds for all $r \times r$ matrices X of real numbers, then it also holds when X is a matrix of indeterminate. Thus $\mathrm{Tr}(X)$ and $\det(X)$ are invariant polynomials.

Example 4.2. Let $\det(\lambda I + X) = \lambda^r + f_1(X)\lambda^{r-1} + \cdots + f_r(X)$, then $f_k(X)$ is an invariant polynomial of degree k. In fact, $(-1)^k f_k(X)$ is the k-th coefficient of the characteristic polynomial of X.

Example 4.3. Let $\Sigma_k(X) = \mathrm{Tr}(X^k)$ be the trace of X^k, then $\Sigma_k(X)$ is an invariant polynomial of degree k.

Proposition 4.4. *Let $\Phi(X)$ be an invariant polynomial on $\mathfrak{gl}(r, \mathbb{R})$. Then $\Phi(\Omega_\alpha)$ and $\Phi(\Omega_\beta)$ coincide on $U_\alpha \cap U_\beta$, thus determine a global form $\Phi(R^\nabla)$ on M.*

Proof. Since $\Omega_\beta = g_{\alpha\beta}^{-1}\Omega_\alpha g_{\alpha\beta}$, we have

$$\Phi(\Omega_\beta) = \Phi(g_{\alpha\beta}^{-1}\Omega_\alpha g_{\alpha\beta}) = \Phi(\Omega_\alpha), \quad \text{on } U_\alpha \cap U_\beta. \qquad \square$$

Proposition 4.5. *The global form $\Phi(R^\nabla)$ is closed.*

To prove this proposition, we need the following fact about invariant polynomials.

Fact 2. The algebra of invariant polynomials on $\mathfrak{gl}(r, \mathbb{R})$ is the polynomial algebra generated by $\Sigma_1, \ldots, \Sigma_r$ (or equivalently by $f_1, \ldots, f_r$), i.e.,

$$\mathrm{Inv}(\mathfrak{gl}(r, \mathbb{R})) = \mathbb{R}[\Sigma_1, \ldots, \Sigma_r] = \mathbb{R}[f_1, \ldots, f_r].$$

For example, suppose Φ is a homogeneous invariant polynomial of degree 3 on $\mathfrak{gl}(3, \mathbb{R})$, then Φ must be a linear combination of Σ_1^3, $\Sigma_1\Sigma_2$ and Σ_3, i.e.,

$$\Phi(X) = a\Sigma_1(X)^3 + b\Sigma_1(X)\Sigma_2(X) + c\Sigma_3(X), \qquad a, b, c \in \mathbb{R}.$$

Proof of Proposition 4.5. Since Φ is a polynomial in $\Sigma_1, \ldots, \Sigma_r$, it suffices to show that $\mathrm{d}\mathrm{Tr}(R^\nabla) = 0$ for all k. We have

$$\mathrm{d}\mathrm{Tr}(\Omega_\alpha^k) = \mathrm{Tr}(\mathrm{d}\Omega_\alpha^k) = \mathrm{Tr}(\Omega_\alpha^k \wedge \omega_\alpha - \omega_\alpha \wedge \Omega_\alpha^k) = 0. \qquad \square$$

Proposition 4.6. *The class $[\Phi(R^\nabla)]$ is independent of the choice of ∇. It is a (differential) topological invariant of E.*

Proof. For the same reason as above, it suffices to show that $[\mathrm{Tr}(R^\nabla)]$ is independent of the choice of ∇. Let ∇_0, ∇_1 be two connections on E, then $\nabla_t = (1-t)\nabla_0 + t\nabla_1$ is a family of connections on E, whose connection form is given by

$$\omega_t = (1-t)\omega_0 + t\omega_1,$$

and curvature form is

$$\Omega_t = \mathrm{d}\omega_t + \omega_t \wedge \omega_t.$$

So

$$\frac{\mathrm{d}\Omega_t}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}(\mathrm{d}\omega_t + \omega_t \wedge \omega_t) = \frac{\mathrm{d}}{\mathrm{d}t}(\mathrm{d}\omega_t) + \frac{\mathrm{d}\omega_t}{\mathrm{d}t} \wedge \omega_t + \omega_t \wedge \frac{\mathrm{d}\omega_t}{\mathrm{d}t}$$

Then

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{Tr}(\Omega_t^k) &= \mathrm{Tr}\left(\frac{\mathrm{d}}{\mathrm{d}t}\Omega_t^k\right) = \mathrm{Tr}\left(\frac{\mathrm{d}\Omega_t}{\mathrm{d}t}\wedge\Omega_t^{k-1}+\cdots+\Omega_t^{k-1}\wedge\frac{\mathrm{d}\Omega_t}{\mathrm{d}t}\right) = k\mathrm{Tr}\left(\frac{\mathrm{d}\Omega_t}{\mathrm{d}t}\wedge\Omega_t^{k-1}\right)\\
&= k\mathrm{Tr}\left(\left(\mathrm{d}\frac{\mathrm{d}\omega_t}{\mathrm{d}t}+\frac{\mathrm{d}\omega_t}{\mathrm{d}t}\wedge\omega_t+\omega_t\wedge\frac{\mathrm{d}\omega_t}{\mathrm{d}t}\right)\wedge\Omega_t^{k-1}\right)\\
&= k\mathrm{Tr}\left(\mathrm{d}\left(\frac{\mathrm{d}\omega_t}{\mathrm{d}t}\right)\wedge\Omega_t^{k-1}-\frac{\mathrm{d}\omega_t}{\mathrm{d}t}\wedge(\Omega_t^{k-1}\wedge\omega_t-\omega_t\wedge\Omega_t^{k-1})\right)\\
&= k\mathrm{Tr}\left(\mathrm{d}\left(\frac{\mathrm{d}\omega_t}{\mathrm{d}t}\right)\wedge\Omega_t^{k-1}-\frac{\mathrm{d}\omega_t}{\mathrm{d}t}\wedge\mathrm{d}\Omega_t^{k-1}\right)\\
&= k\mathrm{Tr}\left(\mathrm{d}\left(\frac{\mathrm{d}\omega_t}{\mathrm{d}t}\wedge\Omega_t^{k-1}\right)\right)\\
&= \mathrm{d}\,k\mathrm{Tr}\left(\frac{\mathrm{d}\omega_t}{\mathrm{d}t}\wedge\Omega_t^{k-1}\right).
\end{aligned}$$

Therefore,

$$\begin{aligned}
\mathrm{Tr}(\Omega_1^k)-\mathrm{Tr}(\Omega_0^k) &= \int_0^1\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{Tr}(\Omega_t^k)\mathrm{d}t = \int_0^1\mathrm{d}\,k\mathrm{Tr}\left(\frac{\mathrm{d}\omega_t}{\mathrm{d}t}\wedge\Omega_t^{k-1}\right)\mathrm{d}t\\
&= \mathrm{d}\int_0^1 k\mathrm{Tr}\left(\frac{\mathrm{d}\omega_t}{\mathrm{d}t}\wedge\Omega_t^{k-1}\right)\mathrm{d}t. \qquad (4.2)
\end{aligned}$$

Notice that

$$\omega_{t,\beta}=g_{\alpha\beta}^{-1}\omega_{t,\alpha}g_{\alpha\beta}+g_{\alpha\beta}^{-1}dg_{\alpha\beta}\implies\frac{\mathrm{d}\omega_{t,\beta}}{\mathrm{d}t}=g_{\alpha\beta}^{-1}\frac{\mathrm{d}\omega_{t,\alpha}}{\mathrm{d}t}g_{\alpha\beta},\quad\Omega_{t,\beta}=g_{\alpha\beta}^{-1}\Omega_{t,\alpha}g_{\alpha\beta}.$$

So $\mathrm{Tr}\left(\frac{\mathrm{d}\omega_t}{\mathrm{d}t}\wedge\Omega_t^{k-1}\right)$ defines a global form on M. By (4.2), we have $[\mathrm{Tr}(\Omega_1^k)]=[\mathrm{Tr}(\Omega_0^k)]$.

For a general invariant polynomial Φ, we can express Φ as a linear combination of monomials $\Sigma_1^{k_1}\cdots\Sigma_r^{k_r}$. Then $\Phi(\Omega_t)$ is a linear combination of $\Sigma_1^{k_1}(\Omega_t)\wedge\cdots\wedge\Sigma_r^{k_r}(\Omega_t)$. Thus $[\Phi(\Omega_1)]=[\Phi(\Omega_t)]=[\Phi(\Omega_0)]$. □

Definition 4.7. By above propositions, we define the **Chern-Weil homomorphism** as

$$\begin{aligned}
c_E:\mathrm{Inv}(\mathfrak{gl}(r,\mathbb{R}))&\to H_{\mathrm{dR}}^*(M),\\
\Phi&\mapsto[\Phi(R^\nabla)].
\end{aligned}$$

This map sends $\Phi(X)\Psi(X)$ to $[\Phi(R^\nabla)\wedge\Psi(R^\nabla)]=[\Phi(R^\nabla)]\cdot[\Psi(R^\nabla)]$, thus is an algebra homomorphism. This map does not depend on the choice of ∇.

Definition 4.8. A **characteristic class** of a vector bundle $\pi:E\to M$ is a map

$$c:\mathrm{Vect}_k(M)\to H_{\mathrm{dR}}^*(M),\qquad E\mapsto c(E),$$

such that if $f:N\to M$ is a smooth map, then

$$c(f^*E)=f^*c(E).$$

In functional language, a **characteristic class** is a natural transformation from the functor $\mathrm{Vect}_k(-)$ to $H^*_{\mathrm{dR}}(-)$. The naturality condition can be visualized by the following commutative diagram:

$$\begin{array}{ccc} \mathrm{Vect}_k(M) & \xrightarrow{c} & H^*_{\mathrm{dR}}(M) \\ {\scriptstyle f^*}\uparrow & & {\scriptstyle f^*}\uparrow \\ \mathrm{Vect}_k(N) & \xrightarrow{c} & H^*_{\mathrm{dR}}(N) \end{array}$$

Naturality of Chern-Weil homomorphism

Suppose $\pi : E \to M$ is a rank r vector bundle, ∇ a connection on E. Let $f : N \to M$ be a smooth map, it induces a pullback bundle $f^*E \to N$ and a pullback connection $f^*\nabla$ on f^*E

$$(f^*\nabla)(f^*s) := f^*(\nabla s), \qquad \forall\, s \in \Gamma(E).$$

Suppose ω is the connection form of ∇, then $f^*\omega$ is the connection form of $f^*\nabla$. The curvature form of $f^*\nabla$ is therefore $f^*(\mathrm{d}\omega + \omega \wedge \omega) = f^*\Omega$. For any invariant polynomial Φ, we have

$$\Phi(f^*\Omega) = f^*\Phi(\Omega) \implies c_{f^*E}(\Phi) = [\Phi(f^*\Omega)] = f^*[\Phi(\Omega)] = f^*c_E(\Phi).$$

which shows the naturality of Chern-Weil homomorphism.

4.2 Examples of Characteristic Classes

We have introduced a powerful mechinery to construct characteristic classes of vector bundles. In this subsection, we will give a few examples of characteristic classes.

4.2.1 Pontryagin Classes

In this subsection, we only consider real vector bundles. Recall that

$$\det(\lambda I + X) = \lambda^r + f_1(X)\lambda^{r-1} + \cdots + f_r(X),$$

in particular, $f_1(X) = \mathrm{Tr}(X)$. If X is a skew-symmetric matrix, then $f_1(X) = 0$. In general we have the following proposition.

Proposition 4.9. *If Ω is skew-symmetric, then $\mathrm{Tr}(\Omega^k) = 0$ for all odd k. Moreover, $\Phi(\Omega) = 0$ for all invariant polynomials Φ of odd degree.*

Proof. Since Ω is skew-symmetric,

$$(\Omega \wedge \cdots \wedge \Omega)^T = (\Omega^T) \wedge \cdots \wedge (\Omega^T) = (-\Omega) \wedge \cdots \wedge (-\Omega) = (-1)^k \Omega \wedge \cdots \wedge \Omega.$$

So Ω^k is skew-symmetric for odd k, and $\mathrm{Tr}(\Omega^k) = 0$.

For a general invariant polynomial Φ of odd degree, since Φ is a linear combination of monomials $\Sigma_1^{k_1} \cdots \Sigma_r^{k_r}$ with $k_1 + 2k_2 + \cdots + rk_r$ odd, at least one of $k_1, k_3, \ldots$ is odd, thus $\Phi(\Omega)$ must be zero. $\square$

Since $[\Phi(R^\nabla)]$ is independent of the choice of ∇, we can choose a special connection ∇ such that R^∇ is skew-symmetric. To do this, we choose a Riemannian metric on E, then there exists connections on E that are compatible with the metric. Under a local orthonormal frame, the connection form ω is skew-symmetric, so as the curvature form Ω. By the above proposition, $[\Phi(R^\nabla)] = 0$ for all invariant polynomials Φ of odd degree.

Corollary 4.10. *For a real vector bundle E, the ring of characteristic classes of E has two sets of generators:*

$$[\mathrm{Tr}(\Omega^2)],\ [\mathrm{Tr}(\Omega^4)],\ [\mathrm{Tr}(\Omega^6)], \dots;$$

and

$$[f_2(\Omega)],\ [f_4(\Omega)],\ [f_6(\Omega)], \dots.$$

Definition 4.11. The k-th **Pontryagin class** of a real vector bundle E is defined to be

$$p_k(E) := (-1)^k \left[f_{2k} \left(\frac{1}{2\pi} R^\nabla \right) \right] = \left[f_{2k} \left(\frac{\sqrt{-1}}{2\pi} R^\nabla \right) \right] \in H^{4k}(M).$$

Remark. The factor $\sqrt{-1}$ is for the sign convention. The factor $\frac{1}{2\pi}$ is for the integrality of Pontryagin classes. A class $\alpha \in H^q(M)$ is called **integral** if $\int_S \alpha \in \mathbb{Z}$ for all q-dimensional closed oriented submanifolds S of M.

By definition,

$$\begin{aligned} \det \left(\lambda I + \frac{\sqrt{-1}}{2\pi} R^\nabla \right) &= \lambda^r + f_1 \left(\frac{\sqrt{-1}}{2\pi} R^\nabla \right) \lambda^{r-1} + \cdots + f_r \left(\frac{\sqrt{-1}}{2\pi} R^\nabla \right) \\ &= \lambda^r + p_1(E)\lambda^{r-2} + \cdots + p_{[r/2]}(E)\lambda^{r-2[r/2]}. \end{aligned}$$

Apply $\lambda = 1$,

$$\det \left(I + \frac{\sqrt{-1}}{2\pi} R^\nabla \right) = 1 + p_1(E) + \cdots + p_{[r/2]}(E) =: p(E).$$

We call $p(E)$ the **total Pontryagin class** of E.

Proposition 4.12 (Whitney product formula)**.** *For two real vector bundles E and E', we have*

$$p(E \oplus E') = p(E) \cdot p(E').$$

Proof. Let ∇ and ∇' be connections on E and E', then $\nabla \oplus \nabla'$ is a connection on $E \oplus E'$. Under a suitable local trivialization, the curvature form of $\nabla \oplus \nabla'$ is given by

$$F_{\nabla \oplus \nabla'} = \begin{pmatrix} R^\nabla & 0 \\ 0 & R^{\nabla'} \end{pmatrix} \implies \Omega_{E \oplus E'} = \begin{pmatrix} \Omega & 0 \\ 0 & \Omega' \end{pmatrix}.$$

So

$$\begin{aligned} p(E \oplus E') &= \det \left(I + \frac{\sqrt{-1}}{2\pi} \Omega_{E \oplus E'} \right) \\ &= \det \begin{pmatrix} I + \frac{\sqrt{-1}}{2\pi}\Omega & \\ & I + \frac{\sqrt{-1}}{2\pi}\Omega' \end{pmatrix} \\ &= \det \left(I + \frac{\sqrt{-1}}{2\pi}\Omega \right) \cdot \det \left(I + \frac{\sqrt{-1}}{2\pi}\Omega' \right) \\ &= p(E) \cdot p(E'). \end{aligned}$$

□

4.2.2 Euler Classes

If in addition E is orientable and we choose an orientation of E. Then we can make the local orthonormal frame $e = [e_1, \ldots, e_r]$ of E to be compatible with the orientation, so the transition function between two frames is in $\mathrm{SO}(r)$ and the connection form ω is skew-symmetric. This inspires us to consider the $\mathrm{SO}(r)$-invariant polynomial on $\mathfrak{so}(r)$.

Fact 3. For r odd, $\mathrm{Inv}(\mathfrak{so}(r))$ is generated by $\Sigma_1, \ldots, \Sigma_r$ (or equivalently by $f_1, \ldots, f_r$). For r even, $\mathrm{Inv}(\mathfrak{so}(r))$ has one more generator Pf, the **Pfaffian**. That is

$$\begin{aligned}\mathrm{Inv}(\mathfrak{so}(2m+1)) &= \mathbb{R}[\Sigma_1, \ldots, \Sigma_{2m+1}] = \mathbb{R}[f_1, \ldots, f_{2m+1}],\\ \mathrm{Inv}(\mathfrak{so}(2m)) &= \mathbb{R}[\Sigma_1, \ldots, \Sigma_{2m}, \mathrm{Pf}] = \mathbb{R}[f_1, \ldots, f_{2m}, \mathrm{Pf}].\end{aligned}$$

The Pfaffian of skew-symmetric matrices

Let's begin with some examples. For a 2×2 skew-symmetric matrix

$$X = \begin{pmatrix} 0 & a \\ -a & 0 \end{pmatrix},$$

we have

$$\det(X) = a^2.$$

For a 4×4 skew-symmetric matrix

$$X = \begin{pmatrix} 0 & a & b & c \\ -a & 0 & d & e \\ -b & -d & 0 & f \\ -c & -e & -f & 0 \end{pmatrix},$$

after a direct calculation,

$$\det(X) = (af - be + cd)^2.$$

One can observe that both determinants are perfect squares. This phenomenon holds for all skew-symmetric matrices:

Fact 4. For any $2m \times 2m$ skew-symmetric matrix $X = (x^i_j)$, $\det(X)$ is a perfect square in $\mathbb{Z}[x^i_j]$. The square root of $\det(X)$ is called the **Pfaffian** of X, denoted by $\mathrm{Pf}(X)$.

Remark. Notice that the square roots of $\det(X)$ are $\pm\mathrm{Pf}(X)$. To determine the sign, we require $\mathrm{Pf}(J_{2m}) = 1$, where

$$J_{2m} = \begin{pmatrix} 0 & 1 & & & \\ -1 & 0 & & & \\ & & \ddots & & \\ & & & 0 & 1 \\ & & & -1 & 0 \end{pmatrix}.$$

Fact 5. Let $A = (a^i_j)$ and $X = (x^i_j)$ be two $2m \times 2m$ matrices with X skew-symmetric, then

$$\mathrm{Pf}(A^T X A) = \det(A)\mathrm{Pf}(X).$$

If $A \in \mathrm{SO}(2m)$, then $\mathrm{Pf}(A^T X A) = \mathrm{Pf}(X)$, so Pf is an invariant polynomial on $\mathfrak{so}(2m)$.

Definition 4.13. For a $2m$-dimensional oriented real vector bundle $E \to M$, choose a metric on E and a compatible connection ∇, the **Euler class** of E is defined to be

$$e(E) := \left[\mathrm{Pf}\left(\frac{1}{2\pi}R^{\nabla}\right)\right] \in H^{2m}(M).$$

Remark. The Euler class is a characteristic class of oriented real vector bundles. It depends on the orientation: if we reverse the orientation, then $e(E)$ changes to $-e(E)$.

Remark. The definition of Euler class requires a metric and a compatible connection, but the resulting class does not depend on the choice of metric and connection:

- Fixing a metric g on E, any two metric-compatible connections ∇_0 and ∇_1 define the same Euler class.
- Given any two metrics g_0 and g_1 on E, we can choose a family of connections ∇_t such that ∇_0 (resp. ∇_1) is compatible with g_0 (resp. g_1). The whole family ∇_t defines the same Euler class.

Theorem 4.14 (Gauss-Bonnet-Chern theorem)**.** *Let M be a closed oriented Riemannian manifold of dimension $2m$, ∇ be a metric-compatible connection on TM, then*

$$\int_M e(TM) = \int_M \mathrm{Pf}\left(\frac{1}{2\pi}R^{\nabla}\right) = \chi(M),$$

where $\chi(M)$ is the Euler characteristic of M.

Example 4.15. When M is a closed oriented surface, the above theorem reduces to the classical Gauss-Bonnet theorem: let e_1, e_2 be a local orthonormal frame of TM, θ^1, θ^2 be the dual coframe, then the connection and curvature forms are

$$\omega = \begin{pmatrix} 0 & \omega_2^1 \\ -\omega_2^1 & 0 \end{pmatrix}, \qquad \Omega = \mathrm{d}\omega + \omega \wedge \omega = \begin{pmatrix} 0 & \mathrm{d}\omega_2^1 \\ -\mathrm{d}\omega_2^1 & 0 \end{pmatrix},$$

and $\mathrm{d}\omega_2^1 = K\theta^1 \wedge \theta^2$, where K is the Gaussian curvature of M. So

$$\mathrm{Pf}\,(\Omega) = \mathrm{d}\omega_2^1 = K\theta^1 \wedge \theta^2.$$

Therefore,

$$\int_M e(TM) = \frac{1}{2\pi}\int_M K\theta^1 \wedge \theta^2 = \chi(M) = 2 - 2g(M).$$

4.2.3 Chern Classes

For a complex vector bundle $E \to M$ with $\mathrm{rk}_{\mathbb{C}}(E) = r$, we can choose a local trivialization $\{(U_\alpha, \psi_\alpha)\}$ of E such that the transition function $g_{\alpha\beta} \in \mathrm{GL}(r, \mathbb{C})$. Then any complex polynomial Φ on $\mathfrak{gl}(r, \mathbb{C})$ invariant under conjugation by $\mathrm{GL}(r, \mathbb{C})$ defines a characteristic class $[\Phi(R^{\nabla})]$ which is independent of the choice of ∇.

Definition 4.16. The k-th **Chern class** of a complex vector bundle E is defined to be

$$c_k(E) := \left[f_k \left(\frac{\sqrt{-1}}{2\pi} R^\nabla \right) \right] \in H^{2k}(M).$$

The **total Chern class** of E is defined to be

$$c(E) := \det \left(I + \frac{\sqrt{-1}}{2\pi} R^\nabla \right) = 1 + c_1(E) + \cdots + c_r(E).$$

Proposition 4.17. *Chern classes satisfy the naturality property and the Whitney product formula, i.e.,*

$$c(f^*E) = f^*c(E), \qquad c(E \oplus E') = c(E) \cdot c(E').$$

Remark. If we choose a Hermitian metric on E and a compatible connection ∇, then the curvature form R^∇ is skew-Hermitian, so $\frac{\sqrt{-1}}{2\pi} R^\nabla$ is Hermitian. Therefore, $\Sigma_k \left(\frac{\sqrt{-1}}{2\pi} R^\nabla \right) = \operatorname{Tr} \left(\left(\frac{\sqrt{-1}}{2\pi} R^\nabla \right)^k \right)$ is a real form, and $c_k(E)$ is a real cohomology class. In fact, $c_k(E)$ is an integral cohomology class, i.e., $c_k(E) \in H^{2k}(M; \mathbb{Z})$.

Proposition 4.18. *Let E be a complex vector bundle, $\mathrm{rk}_{\mathbb{C}}(E) = r$, $E_{\mathbb{R}}$ be the underlying real vector bundle of E. $E_{\mathbb{R}}$ is oriented and if $e = [e_1, \cdots, e_r]$ is a local frame of E, then $e_{\mathbb{R}} = [e_1, \sqrt{-1}e_1, \cdots, e_r, \sqrt{-1}e_r]$ is a local oriented frame of $E_{\mathbb{R}}$. We set the orientation determined by $e_{\mathbb{R}}$ to be the orientation of $E_{\mathbb{R}}$, then the top Chern class $c_r(E)$ is equal to the Euler class $e(E_{\mathbb{R}})$.*

Proof. Choose a Hermitian metric $\langle \cdot, \cdot \rangle$ on E and a compatible connection ∇, then the curvature R^∇ is skew-Hermitian. One can define a Riemannian metric on $E_{\mathbb{R}}$ by

$$\langle u, v \rangle_{\mathbb{R}} := \operatorname{Re}\langle u, v \rangle \qquad \forall\, u, v \in E_{\mathbb{R}}.$$

Then

$$X\langle u, v \rangle_{\mathbb{R}} = \operatorname{Re}\Big(X\langle u, v \rangle \Big) = \operatorname{Re}\Big(\langle \nabla_X u, v \rangle + \langle u, \nabla_X v \rangle \Big) = \langle \nabla_X u, v \rangle_{\mathbb{R}} + \langle u, \nabla_X v \rangle_{\mathbb{R}}.$$

So ∇ is compatible with $\langle \cdot, \cdot \rangle_{\mathbb{R}}$, and the curvature $(R^\nabla)_{\mathbb{R}}$ is skew-symmetric.

Under a local unitary frame $e = [e_1, \ldots, e_r]$ of E, the curvature form Ω is skew-Hermitian, and so $\sqrt{-1}\Omega$ is Hermitian. The frame $e_{\mathbb{R}} = [e_1, \sqrt{-1}e_1, \cdots, e_r, \sqrt{-1}e_r]$ is a local orthonormal frame of $E_{\mathbb{R}}$, and the curvature form under $e_{\mathbb{R}}$ is skew-symmetric.

If $\Omega = (z^i_j) = (x^i_j + \sqrt{-1}y^i_j)$ with $x^i_j, y^i_j \in \mathbb{R}$, then $\Omega = A + \sqrt{-1}B$ with $A = (x^i_j)$ and $B = (y^i_j)$. Then the curvature form under $e_{\mathbb{R}}$ is given by

$$\Omega_{\mathbb{R}} = \begin{pmatrix} x^1_1 & -y^1_1 & \cdots & x^1_r & -y^1_r \\ y^1_1 & x^1_1 & \cdots & y^1_r & x^1_r \\ \vdots & \vdots & & \vdots & \vdots \\ x^r_1 & -y^r_1 & \cdots & x^r_r & -y^r_r \\ y^r_1 & x^r_1 & \cdots & y^r_r & x^r_r \end{pmatrix} \underset{\rightsquigarrow}{\text{similar to}} \begin{pmatrix} x^1_1 & \cdots & x^1_r & -y^1_1 & \cdots & -y^1_r \\ \vdots & & \vdots & \vdots & & \vdots \\ x^r_1 & \cdots & x^r_r & -y^r_1 & \cdots & -y^r_r \\ y^1_1 & \cdots & y^1_r & x^1_1 & \cdots & x^1_r \\ \vdots & & \vdots & \vdots & & \vdots \\ y^r_1 & \cdots & y^r_r & x^r_1 & \cdots & x^r_r \end{pmatrix} = \begin{pmatrix} A & -B \\ B & A \end{pmatrix}$$

So

$$\det\left(\frac{1}{2\pi}\Omega_{\mathbb{R}}\right) = \det\left(\frac{1}{2\pi}\begin{pmatrix} A & -B \\ B & A \end{pmatrix}\right) = \frac{1}{(2\pi)^{2r}}\left|\det(A+\sqrt{-1}B)\right|^2$$
$$= \left|\det\left(\frac{1}{2\pi}\Omega\right)\right|^2 = \left|\det\left(\frac{\sqrt{-1}}{2\pi}\Omega\right)\right|^2 = \left(\det\left(\frac{\sqrt{-1}}{2\pi}\Omega\right)\right)^2.$$

The last equality holds since $\det\left(\frac{\sqrt{-1}}{2\pi}\Omega\right)$ is a real form. Therefore,

$$\mathrm{Pf}\left(\frac{1}{2\pi}(R^{\nabla})_{\mathbb{R}}\right) = \pm\det\left(\frac{\sqrt{-1}}{2\pi}R^{\nabla}\right).$$

To determine the sign, set $\Omega = -\sqrt{-1}I_r$, then

$$\Omega_{\mathbb{R}} = \begin{pmatrix} 0 & 1 & & & \\ -1 & 0 & & & \\ & & \ddots & & \\ & & & 0 & 1 \\ & & & -1 & 0 \end{pmatrix} = J_{2r}$$

So

$$\det(\sqrt{-1}\Omega) = \det(I_r) = 1 = \mathrm{Pf}(J_{2r}) = \mathrm{Pf}(\Omega_{\mathbb{R}}).$$

The sign under this setting is $+1$, so $c_r(E) = e(E_{\mathbb{R}})$. □

4.3 Characteristic classes of Principal Bundles